

HOMESTEAD AFB, FL, USA

WMO: 722026

Lat: 25.483N

Lon: 80.383W

Elev: 2

StdP: 101.31

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 12826

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.9	10.1	1.0	4.1	12.1	3.9	5.0	12.7	10.6	19.4	9.2	19.9	3.0	340	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	32.9	26.3	32.5	26.2	32.1	26.1	27.5	30.8	27.2	30.5	26.8	30.3	4.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.6	29.2	26.2	21.7	28.8	25.9	21.2	28.7	87.1	31.1	85.0	30.5	83.3	30.6	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	8.1	7.4	4.5	34.3	2.3	0.9	2.8	35.0	1.5	35.5	0.3	36.0	-1.4	36.7		
			2.8	28.1	2.8	0.6	0.9	28.5	-0.7	28.9	-2.3	29.2	-4.3	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.6	19.4	20.9	22.2	24.0	25.9	27.6	28.3	28.4	27.8	26.1	23.1	21.3	(6)
(7)		DBStd	3.88	3.86	3.41	2.95	2.17	1.43	1.16	1.06	0.98	1.03	1.99	2.61	3.60	(7)
(8)		HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	80	36	16	8	1	0	0	0	0	0	0	3	17	(9)
(10)		CDD10.0	5334	293	306	379	421	494	529	566	571	534	498	393	351	(10)
(11)		CDD18.3	2372	70	88	128	171	236	279	308	312	284	240	145	110	(11)
(12)		CDH23.3	21277	282	441	744	1212	2063	2927	3496	3577	2982	2146	873	533	(12)
(13)	CDH26.7	6634	13	44	126	255	567	1064	1391	1418	1028	562	122	44	(13)	
(14)	Wind	WSAvg	3.3	3.5	3.4	3.9	3.9	3.7	2.8	2.6	2.6	2.9	3.7	3.6	3.3	(14)
(15)	Precipitation	PrecAvg	1475	48	52	54	72	135	239	175	212	215	155	63	54	(15)
(16)		PrecMax	1838	152	147	147	139	318	472	275	313	334	371	202	314	(16)
(17)		PrecMin	1222	10	1	7	8	14	78	93	83	95	38	10	8	(17)
(18)		PrecStd	174	34	35	34	35	72	106	54	60	63	92	56	63	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	29.1	30.6	31.5	32.4	33.7	33.9	33.6	33.0	32.0	30.2	28.8	(19)
(20)			MCWB	22.9	23.2	23.4	24.0	25.1	26.5	26.5	26.5	25.9	25.6	24.8	24.3	(20)
(21)		2%	DB	27.1	27.9	29.1	30.2	31.2	32.6	32.8	32.8	32.3	31.0	29.0	27.8	(21)
(22)			MCWB	22.5	22.9	23.3	23.6	24.9	26.2	26.3	26.5	26.0	25.6	24.4	23.6	(22)
(23)		5%	DB	26.1	27.1	27.9	29.1	30.2	31.9	32.3	32.3	31.5	30.2	27.9	27.2	(23)
(24)			MCWB	22.0	22.5	22.7	23.3	24.5	26.0	26.2	26.4	25.9	25.3	23.5	23.2	(24)
(25)	10%	DB	25.1	26.1	27.2	28.0	29.5	31.1	31.7	31.8	31.0	29.6	27.2	26.2	(25)	
(26)		MCWB	21.2	21.9	22.2	22.7	24.3	25.7	26.0	26.2	25.9	25.1	22.8	22.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.8	25.0	25.9	25.7	26.9	28.2	28.0	28.1	27.5	27.3	26.0	25.6	(27)
(28)			MADB	26.3	27.0	27.9	29.2	29.8	31.2	31.6	31.7	30.3	29.5	27.9	26.9	(28)
(29)		2%	WB	23.6	24.0	24.3	24.9	25.9	27.4	27.3	27.4	27.1	26.5	25.2	24.6	(29)
(30)			MADB	25.6	26.3	27.6	28.4	29.2	30.5	31.1	30.9	30.1	29.4	27.8	26.5	(30)
(31)		5%	WB	22.8	23.4	23.6	24.2	25.4	26.8	26.9	26.9	26.7	26.0	24.5	24.0	(31)
(32)			MADB	25.2	25.9	26.7	27.6	28.9	30.2	30.7	30.6	29.9	28.9	27.1	26.1	(32)
(33)	10%	WB	22.0	22.7	22.8	23.6	24.8	26.2	26.5	26.6	26.3	25.6	23.9	23.3	(33)	
(34)		MADB	24.3	25.4	25.9	27.0	28.3	29.8	30.4	30.4	29.6	28.5	26.4	25.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	9.0	8.7	8.4	7.8	7.1	7.0	7.0	6.9	6.5	6.4	7.6	7.9	(35)	
(36)		5% DB	MADB	7.9	8.4	8.9	8.9	8.4	8.1	7.7	7.5	7.2	6.6	7.0	7.0	(36)
(37)			MCWBR	4.7	4.5	4.3	4.1	3.9	3.8	3.4	3.2	3.0	2.9	3.5	3.9	(37)
(38)		5% WB	MADB	7.3	7.6	8.2	7.6	7.6	7.4	7.1	7.1	6.6	6.1	6.4	6.3	(38)
(39)			MCWBR	4.8	4.3	4.5	4.0	3.9	3.7	3.6	3.3	3.1	3.0	3.4	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.363	0.378	0.387	0.399	0.427	0.457	0.464	0.452	0.432	0.417	0.382	0.375	(40)	
(41)		taud	2.536	2.503	2.465	2.415	2.383	2.360	2.331	2.394	2.442	2.452	2.500	2.519	(41)	
(42)		Ebn at Noon	892	904	910	901	869	838	832	843	855	851	863	864	(42)	
(43)		Edh at Noon	92	102	111	119	123	125	128	120	112	106	95	91	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.76	4.53	5.55	6.25	6.25	5.68	5.86	5.65	4.98	4.58	3.95	3.50	(44)	
(45)		RadStd	0.24	0.25	0.24	0.29	0.49	0.49	0.23	0.29	0.25	0.32	0.18	0.26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	+0.38	+0.88	N/A	N/A	N/A	N/A	N/A	-17	+129	+107

Nomenclature: See separate page